Tanner Dean

tjdean2@illinois.edu | (757) 705-2554 | U.S. Citizen | Available June 2026

EDUCATION

Virginia Tech
Blacksburg, VA

Major: B.S. Biochemistry
Minor: Chemistry
GPA: 3.79

2017–2021

University of Illinois Urbana-Champaign (UIUC)
Urbana, IL

Degree: Ph.D. Biophysics & Quantitative Biology
GPA: 3.97

2021–Present

RESEARCH

Molecular Dynamics Simulations of Nucleic Acids 2018–2021

Project Focus: Characterize structure, dynamics, and stability of G-Quadruplexes with the Drude polarizable force field.

Department of Biochemistry, Virginia Tech

Principal Investigator: Prof. Justin Lemkul

Markov State Models for Analysis of Membrane Transport 2022-Present

Project Focus: Elucidate the mechanism of transport for the bacterial nitrate transporter NarU with a Markov State Model.

Department of Chemical and Biomolecular Engineering, UIUC

Principal Investigator: Prof. Diwakar Shukla

Deep learning Model for Enzyme-Substrate Prediction (EZSpecificity) 2022-2025

Project Focus: Development of a state-of-the-art deep learning model for predicting enzymes for novel substrates utilizing sequence and structural information.

Department of Chemical and Biomolecular Engineering, UIUC

Principal Investigator: Prof. Diwakar Shukla

PUBLICATIONS

Salsbury, A. M., **Dean, T. J.**, & Lemkul, J. A. (2020). Polarizable Molecular Dynamics Simulations of Two c-kit Oncogene Promoter G-Quadruplexes: Effect of Primary and Secondary Structure on Loop and Ion Sampling. *Journal of Chemical Theory and Computation*, *16*(5). https://doi.org/10.1021/acs.jctc.0c00191

Chen, Jiming, **Dean, T. J.**, & Shukla, Diwakar (2024). Contribution of Signaling Partner Association to Strigolactone Receptor Selectivity. *Journal of Physical Chemistry B*, *128(3)*. https://doi.org/10.1021/acs.jpcb.3c06940

Sobecks, B. L., Chen, J., **Dean, T. J.**, & Shukla, D. (2025). Mechanistic Basis for Enhanced Strigolactone Sensitivity in KAI2 Triple Mutant. *Biophysical Journal, 124(12).* https://doi.org/10.1016/j.bpj.2025.04.021

Dean, T. J.†, Clark, J. D.†, & Shukla, D. (2025). Two for the Price of One: Integrating Large Language Models to Learn Biophysical Interactions. *Commun. Chem.* https://doi.org/10.1038/s42004-025-01883-7

Cui, H.†, Su, Y.†, **Dean, T. J.**†, Yu, T., Zhang, Z., Peng, J., Shukla, D., Zhao, H. (2025). Predicting Enzyme Substrate Specificity using Cross Attention-enhanced Graph Neural Networks. *Nature.* https://doi.org/10.1038/s41586-025-09697-2

Dean, T. J., Feng, J., Shukla, D. (2026). How minor sequence changes enable mechanistic diversity in MFS transporters? An atomic-level rationale for symport emergence in NarU. *J. Chem. Inf. Model.* https://doi.org/10.1021/acs.jcim.5c02971

PRESENTATIONS

Structure and Dynamics of the *c-myc* G-Quadruplex

T. Dean, A. Salsbury, J. Lemkul

Poster: *Dennis Dean Undergraduate Research and Creative Scholarship Conference, 2019 Engelpalooza*

Dynamics of the 1:2:1 and 1:6:1 *c-myc* G-Quadrduplexes With the Drude Polarizable Force Field

T. Dean, A. Salsbury, J. Lemkul

Poster: *2020 Annual Meeting of the Biophysical Society*

Predicting the Substrate Specificity and Regioselectivity of Halogenases Using Deep Learning

H. Cui, Y. Su, T. Dean, D. Shukla, H. Zhao

Poster: *2023 AIChE Annual Meeting*

Virtual Workshop: Machine Learning For Molecular Plant Sciences

T. Dean, D. Shukla, G. Moghe

Workshop: *American Society of Plant Biologists (Plant Biology 2024)*

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SERVICE AND OUTREACH

Curie & DaVinci (Orion) Living-Learning Community (2019-2020)

- Mentored several freshman biochemistry majors throughout the school year. Helped to solve questions on university, academics, undergraduate research, and end of the year project as assigned project head.

Folding@home Science & Communications Team (2022-2024)

- Work on a committee to act as public communication between the scientists who use Folding@home and the citizen scientists who donate their computing time to science.

- Plan social media posts, presentations of ongoing science, and community interactions with the public, as well as communications with volunteer beta testers.

CURIE (UIUC ChBE Summer Camp) Volunteer (2022-present)

- Every Summer, introduce high schoolers to computational chemistry through hosting camp activities around VMD.

- Every Summer, updating the activities to keep up with famous proteins in the news like SARS-CoV2 spike protein, GLP-1 receptor (protein target of Ozempic), and insulin.

NSF Molecule Maker Lab Institute Summer Camp and After-School Program Volunteer (2022-2024)

- Helped to develop a learning module and activities on the importance of enzymes and the application of computational chemistry to rational enzyme design.

- Introduced several after-school groups and summer camps to molecular docking, machine learning, and computational modelling.

AWARDS

- **Dean's List (2017-2021)**
- **Archa Vaughan, Jr. Scholarship (Spring 2020-Spring 2021)**
- **COVID-19 Innovation Competition Seed Grant Round Winners, Treasurer and Proposal Writer**

RELEVANT SKILLS AND COURSEWORK

- **General tools-** Linux, Unix, MS Office suite, Vim
- **Proficient in-** Python, C++
- **Familiar with-** CSS, HTML5, R

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- **Professional Software-** NAMD, AMBER, CHARMM, GROMACS, OPENMM, VMD, and PyMOL, PyTorch, TensorFlow
- **Relevant Courses-** General Biochemistry, Drug Chemistry, Physical Chemistry, Bioinformatics, Biophysics, Data Science for Chemistry & Engineering

ORGANIZATIONS AND AFFILIATIONS

Bioactivity (2017-2019)

- Conducted work for biochemistry sub-team in protein-ligand docking and initiating production of a bioprinter project.

Folding@home MSMBuilder Software Development Group (2022-2023)

- Work on maintaining current code and resolving user issues, as well as implementation of more recent algorithms and methods to the existing codebase.

Illinois Biophysics (2022-2025)

- Served as a member of the Executive Committee, Vice President, and President of the organization.

- Presided over planning social events as well as professional symposiums targeted towards members of the Center for Biophysics and local science community at UIUC.